CAMPUS SHUTTLE NAVIGATOR

Eisenhower Matrix — Feature Prioritization

DO — Urgent and Important

Core features for the first version (MVP)

1. Interactive campus map: Display available shuttle stops and routes on a campus map.

2. Nearest-stop finder: Identify nearby stops using browser location, with manual stop selection as a fallback.

3. Destination-based route selection: Match a student's chosen destination to the available shuttle route options.

4. Arrival and journey estimates: Show scheduled or estimated arrival information and approximate journey duration, clearly labelled.

SCHEDULE — Important, Not Urgent

Possible improvements after the MVP

1. Live vehicle location: Add GPS-based shuttle tracking if an authorized and reliable data feed is available.

2. Delay and service notifications: Notify students about verified delays or service changes when an official source is available.

3. Improved arrival predictions: Improve estimates using more complete schedule or historical data.

4. Favourite stops and routes: Allow students to save frequently used stops and journeys.

DELEGATE — Urgent, Not Important

Use existing tools, libraries, and APIs instead of building them from scratch.

1. Leaflet.js: Use the existing library for the interactive map, stop markers, and route display.

2. OpenStreetMap: Use existing map tiles and geographic information for the campus map.

3. Geolocation API: Use the browser API to obtain the student's location with permission.

4. Fetch API and Response.json(): Use built-in browser methods to load and read the local stops and routes data.

DELETE — Not Urgent, Not Important

Features outside the project's intended scope

1. In-app payments: Ticket purchasing or fare processing.

2. User accounts: Registration and profile management.

3. Shuttle reservations: Seat booking or bus reservations.

4. Social features: Messaging, reviews, or community discussions.